

Creating and Analyzing Disk Images Using dc3dd and Autopsy – Working Commands Guide

This PDF contains the corrected and successfully working commands executed in Kali Linux for the digital forensics experiment using dc3dd and Autopsy.

1. Check Disk Partitions

```
lsblk
```

2. Create Sample Evidence File

```
echo "cs evidence" > evidence.txt
```

3. Verify File Creation

```
ls
```

4. Install dc3dd

```
sudo apt install dc3dd
```

5. Create 100MB Disk Image

```
dd if=/dev/zero of=~/practice1.dd bs=1M count=100
```

6. Format Disk Image with EXT4

```
mkfs.ext4 ~/practice1.dd
```

7. Create Mount Directory

```
mkdir ~/mnt
```

8. Mount Disk Image

```
sudo mount -o loop ~/practice1.dd ~/mnt
```

9. Add Evidence into Mounted Image

```
echo "cs di" | sudo tee ~/mnt/evidence1.txt
```

10. Generate MD5 Hash Log

```
sudo sh -c 'md5sum /home/varma/practice1.dd > /home/varma/acquisition.log'
```

11. View Acquisition Log

```
cat /home/varma/acquisition.log
```

12. Start Autopsy

autopsy

Important Notes:

- The filename used successfully was **practice1.dd**.
- The command **mkfs.ext4** is correct; **mxfs.ext4** is invalid.
- dc3dd commands using /dev/zero aborted because of unsupported options in the installed version.
- MD5 hashing was successfully generated manually using md5sum.
- Open Autopsy in the browser using:
<http://localhost:9999/autopsy>